A Project Report

Background Removal and Masking Using Computer Vision

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Abstract

This project focuses on developing a real-time background removal and masking system using computer vision techniques. The system captures live video from a webcam and separates the foreground object from the background dynamically. OpenCV-based image processing methods are used for masking, segmentation, and foreground extraction. The project demonstrates practical implementation of real-time video processing and object isolation techniques.

Objectives

• Develop a real-time background removal system.
• Implement masking techniques using image processing.
• Extract foreground objects dynamically from live video.
• Understand segmentation and background subtraction concepts.
• Build a software-only prototype suitable for future AI integration.

Technologies Used

The project uses the following technologies and tools:
• Python Programming Language
• OpenCV Library
• NumPy
• Webcam / Laptop Camera
• VS Code or PyCharm IDE
• GitHub for version control and project hosting

System Workflow

1. Capture live video frames from the webcam.
2. Apply preprocessing techniques to frames.
3. Detect foreground using background subtraction.
4. Generate masks for separating foreground and background.
5. Remove noise using morphological operations.
6. Replace or remove the background dynamically.
7. Display the processed output in real time.

Implementation Details

The implementation uses OpenCV for live video acquisition and image processing. Background subtraction techniques such as MOG2 are used to detect moving foreground objects. Generated masks are refined using erosion, dilation, and smoothing operations to reduce noise. Bitwise operations are applied to isolate the foreground object while removing or replacing the background. The system performs processing in real time through continuous webcam input.

Applications

- Video Conferencing Background Removal
- Virtual Background Systems
- Security and Surveillance Systems
- AI-based Video Editing
- Human Detection Systems
- Educational Computer Vision Projects

Results

The system successfully removes or masks the background in real time using standard computer vision techniques. Foreground extraction is achieved efficiently with live webcam input. Minor noise and segmentation errors may occur in complex backgrounds, but the prototype demonstrates effective real-time masking and background removal.

Future Scope

Future improvements may include:

- AI-based semantic segmentation.
- Integration with MediaPipe Selfie Segmentation.
- Deep learning-based background removal.
- Green screen replacement support.
- GPU acceleration for faster processing.
- Integration with robotics and AR applications.

Conclusion

This project demonstrates a basic background removal and masking system using computer vision and image processing techniques. It provides a strong foundation for future work in AI-based segmentation, video processing, augmented reality, and intelligent vision systems. The project also helps in understanding practical implementation concepts related to real-time image masking and foreground extraction.

Project Components

Component	Purpose
Python	Main programming language
OpenCV	Image processing and masking
NumPy	Matrix and numerical operations
Webcam	Live video input source
GitHub	Project hosting and version control

GitHub Link

Background_Removal_and_Masking_Project